ChatBS: An Exploratory Sandbox for Bridging Large Language Models with the Open Web

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ABSTRACT

The recent widespread public availability of generative large language models (LLMs) has drawn much attention from the academic community to run experiments in order to learn more about their strengths and drawbacks. From prompt engineering and fine-tuning to fact-checking and task-solving, researchers have pursued several approaches to try to take advantage of these tools. As some of the most powerful LLMs are "closed" and only accessible through web APIs with prior authorization, combining LLMs with the open web is still a challenge. In this evolving landscape, tools that can facilitate the exploration of the capabilities and limitations of LLMs are desirable, especially when connecting with traditional web features such as search and structured data. This article presents ChatBS, a web-based exploratory sandbox for LLMs, working as a front-end for prompting LLMs with user inputs. It provides features such as entity resolution from open knowledge graphs, web search using LLM outputs, as well as popular prompting techniques (e.g. multiple submissions, "step-by-step"). ChatBS has been extensively used in Rensselaer Polytechnic Institute's Data INCITE courses and research, serving as key tool for utilizing LLMs outputs at scale in these contexts.

CCS CONCEPTS

• **Information systems** → **Web applications**; • **Human-centered computing** → Visualization.

KEYWORDS

large language models, fact checking, relationship detection, entity identification

1 INTRODUCTION

Generative large language models (LLMs) have become pervasive in recent years [17], permitting free access to virtually any web user. They are inarguably revolutionary in their ability to generate human-like text with unprecedented scale and speed. Largely led by the industry, these LLMs are trained using vast amounts of web-scraped data, and little is known about exactly what data and how it is used in their training. Some LLMs, such as Meta's Llama [15], are available free of cost (although with terms of usage) and can be downloaded and used (and even fine-tuned). In addition, technical reports are sometimes provided [12], which help users to know a bit more about how the model was originally trained. However, the technical details are often not disclosed as they compose the intellectual property of such companies. Other LLMs, such as OpenAI's GPT family - including ChatGPT, are completely closed and the only way to access them is through web interfaces and APIs. Fine-tuning is sometimes possible, but through the use of APIs as well. The user never has access to the model weights themselves.

This recent proliferation of LLMs has ignited considerable interest within the academic community, prompting researchers to conduct experiments aimed at exploring the strengths and limitations of these powerful tools (e.g. morality [8], kbqa [14]). From innovative techniques like prompt engineering and fine-tuning to their application in task-solving scenarios, a diverse array of approaches has been pursued in a concerted effort to harness the capabilities of LLMs. However, a key challenge arises due to the fact that some of the most formidable LLMs are considered "closed" and are only accessible through web APIs with prior authorization. This presents an obstacle to effectively combining LLMs with the open web, an issue that remains a central concern in the evolving landscape of web science.

In light of this challenge, the development of tools that can facilitate the exploration of the capabilities and limitations of LLMs, especially when integrated with traditional web features like search and structured data, has become increasingly desirable. To address this need, we introduce ChatBS, which serves as a front-end for prompting LLMs with user inputs, offering functionalities to support experimentation and research. On the input side, ChatBS allows users to quickly incorporate popular prompting techniques, such as making multiple submissions and employing a "step-by-step" approach. On the output side, ChatBS includes a web search using LLM-generated content and entity resolution against open knowledge graphs. The latter is accomplished by extracting and linking entities and relationships from the generated responses with the appropriate resources in Wikidata to build an RDF graph. This graph summarizes the LLM's claims in the specific generated content and can be used in conjunction with a Wikidata panel to browse and contrast information from both sources. In this article, we aim to provide an overview of ChatBS, its functionalities, and its impacts, emphasizing its potential to bridge the gap between LLMs and the open web.

2 CHATBS: AN EXPLORATORY SANDBOX

ChatBS was conceived as an accessible, extensible platform for interactive experimentation with LLMs such as GPT 3.5 and GPT 4. It was designed to empower users to easily conduct sophisticated prompt experiments that might be difficult or impossible through the normal ChatGPT user interface, and would normally be done by directly accessing an API (esp. the OpenAI API) programmatically. In addition to providing a variety of experimental options, ChatBS allows for the easy download of experimental results, including model use, prompts, and results, in a machine-readable JSON formatted file.

An important initial use case for ChatBS and the inspiration for its name was fact-checking LLM results. As LLM models became more capable, ChatBS evolved into a useful tool for helping students and researchers replicate experiments from the LLM literature and bootstrap new work.

ChatBS is an R Shiny-based web app that is available to the public at https://inciteprojects.idea.rpi.edu/chatbs/app/chatbs/. We further describe ChatBS's features in the following sections.

2.1 LLM Model Selection

As of Spring 2024, ChatBS enables users to easily select between gpt-4-0613 and gpt-4-turbo-preview models. In the future, OpenAI and other models will be added or dropped depending on availability. The ChatBS team is making plans to diversify model availability by adding the recently-released Llama 3 model [11] as a user option.

2.2 System Prompt Customization

Robust prompt experimentation usually requires more than user prompt manipulation. ChatBS provides a text box for user entry of their desired system prompt, an effective way to pre-define the context, scope, guardrails, or output format for the model to use during an interaction.

The user's custom system prompt is included in the downloadable JSON-formatted results file.

2.3 User Prompt Customization

ChatBS provides a large-capacity text box for customized prompt entry.

2.4 Chain-of-Thought Facilitation: "Explain Step-By-Step"

Important investigations of the reasoning capabilities of LLMs [16] have typically provoked the models under test to enhance their answers or explanations by appending user prompts with *explain step-by-step*. ChatBS provides a checkbox that lets the user append their user prompt with this phrase automatically.

2.5 Explanation Enhancement: "Include References"

A robust and revealing test of an LLM's ability to explain itself, and its limitations, is to request that the LLM document its answers with citations. ChatBS provides a checkbox that lets the user append their user prompt with the phrase "Include references" automatically.

2.6 Repeated Prompt Submission

Experimentation with GPT-3.5 in early 2023 revealed how much variation between responses was typical between repeated prompt submissions. To raise the awareness of users experimenting with LLMs to this natural variation in results, ChatBS allows users to request up to ten re-submissions of their prompts.

We've found that even as the factual correctness of responses has improved with the GPT-4 family of models, the format and contents of specific results across a number of responses will still vary – in other words, many different, correct responses to the same question.

2.7 Relationship Detection, Entity Resolution, and RDF Generation

ChatBS was originally conceived as a compelling and fun way to fact-check LLM-generated answers to questions. The earliest versions of ChatBS (2023) demonstrated the ability to identify asserted relationships between entities within the answers; to resolve entities and relationships (predicates) to URIs; and to express these result graphs as RDF in JSON-LD format. As described elsewhere in this paper, ChatBS uses pre-trained language models for entity and relationship recognition and naive entity linking to establish connection with Wikidata.

2.8 Machine-readable JSON Results Download

In order to document ChatBS experiments and to make the results more shareable and replicable, both the main ChatBS results and the RDF results (if requested) are downloadable as JSON-LD files.

3 SYSTEM ARCHITECTURE

ChatBS was developed to require no installations and configurations by users. As a web application, it is completely accessible through any modern web browser. The ChatBS system architecture comprises four basic elements: UI, OpenAI's continuation API, relationship discovery and entity-linking, and the Wikidata KG. In Figure 1 we highlight and describe the elements of the ChatBS UI. The UI is developed as an R Shiny [5] web application, providing a textual input for users to enter a natural language question about something (just like a user would do when using ChatGPT) and a parameter to set the desired number of answers for the same question (this feature was incorporated to stress the potential different answers, sometimes conflicting, LLMs can generate for the very same prompt). Original responses from the LLM are displayed on the right-hand side panel. Generated triples produced by the relationship discovery module, composing the graph representing the responses' claims, as a table, are optionally displayed by the use of a switch. In addition, a Wikidata snippet conveniently displays the Wikidata web page for any of the entities or relationships linked by the entity-linking module by clicking on any of them. This snippet can be leveraged by users to contrast claims in the triples table against crowd-sourced information in Wikidata, for instance.

Questions submitted by the user are prompted to OpenAI's continuation API. At the time of this writing, ChatBS defaults to using the GPT-4-0613 model. The number of API calls will match the parameter to obtain the desired number of answers.

Figure 1: ChatBS user interface.

3.1 Relationship Discovery and Entity Linking

For each obtained output, ChatBS employs a relationship discovery and entity linking service that parses and analyses the generated content to produce a graph with the LLM's claims. We leverage the approach in [ref]. The current approach involves the use of pre-trained language models for entity and relationship recognition and naive entity linking to establish connection with Wikidata.

4 REFLECTIONS AND IMPACT

ChatBS has proven useful in a variety of ways beyond its initial goal as a LLM fact-checker. As a clean, efficient LLM client optimized for experimentation, it as been especially useful for the *ad hoc* replication of LLM-based research results, for evaluating potential research methods, including prompt engineering strategies, and as a research tool unto itself.

- Wang, et.al. [16] propose a comprehensive trustworthiness evaluation for LLMs that considered diverse perspectives, including toxicity, stereotype bias, adversarial robustness, out-of-distribution robustness, robustness on adversarial demonstrations, privacy, machine ethics, and fairness. ChatBS has proven to be a useful companion to *DecodingTrust*, enabling student users to explore and understand more deeply the paper's many examples.
- Parrish, et.al. [13] propose Bias Benchmark for QA (BBQ), a dataset of question sets that highlight attested social biases against people belonging to protected classes along nine social dimensions relevant for U.S. English-speaking contexts. BBQ evaluates model responses at two levels: (i) given an under-informative context, it tests how strongly responses reflect social biases, and (ii) given an adequately informative context, its test whether the model's biases override a correct answer choice. ChatBS served as test bed for classroom students to build on the *BBQ* work, applying its approach to bias and fairness evaluation to question answering for business and public health dataset.
- LLMs, in particular the GPT-4 family, have proven famously adept at code generation. In the Fall of 2023 ChatBS was used as test bed, and its source code as a launching point, for Rensselaer Data INCITE [2] classroom students to explore the generation of SQL queries and R data transformation pipelines for a very large decentralized finance (DeFi) dataset. This classroom work led to at least one recently submitted publication. [9]
- MortalityMinder is an open-source web-based visualization tool that enables interactive analysis and exploration of social, economic, and geographic factors associated with mortality at the county level. In [4] the authors demonstrate how GPT-4, accessed through ChatBS, can help translate statistical results from MortalityMinder into actionable insights to improve population health. ChatBS helped show that, when combined with MortalityMinder results, GPT-4 provides hypotheses on why socio-economic risk factors are associated with mortality, how they might be causal, and what actions could be taken related to the risk factors to improve outcomes with supporting citations. We find that GPT-4 provided plausible and insightful answers about the relationship between social determinants and mortality.

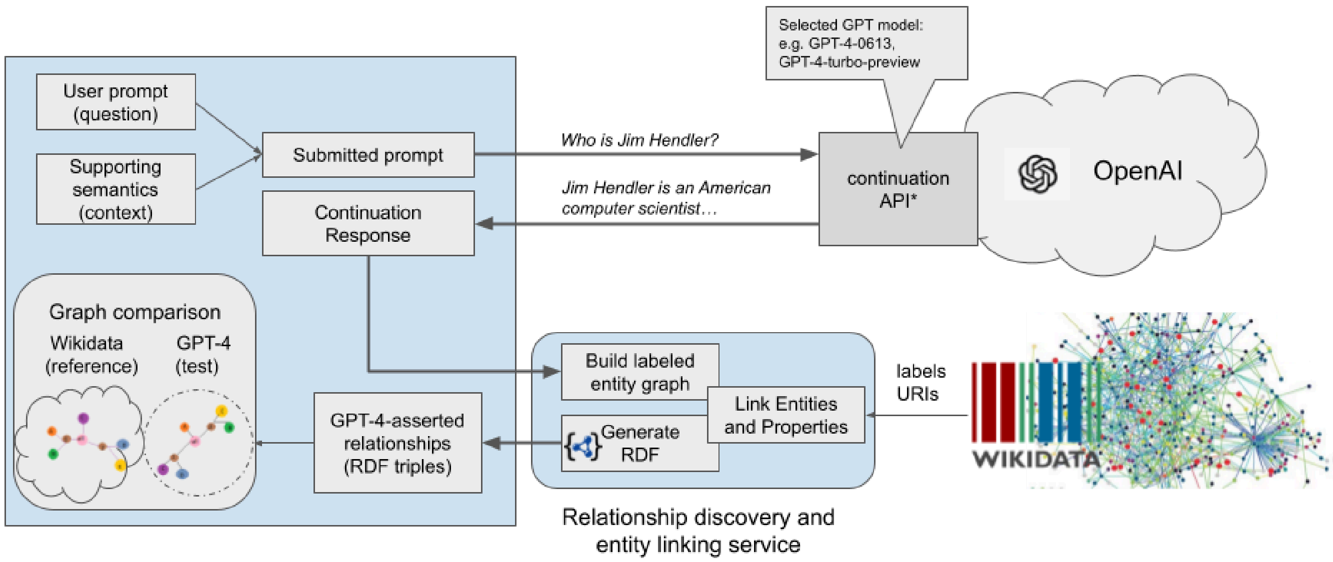

Figure 2: ChatBS architecture.

5 RELATED WORK

At the time ChatBS was first conceived, few resources existed beyond vendor-provided "chat" clients to make robust experimentation with LLMs possible without requiring programmatic access to the APIs. Further, little had been published in early 2023 on the fact checking of LLMs, especially using a knowledge graph-based approach.

Hugging Face [1] is a growing platform on which the machine learning community collaborates on open source LLM models and related datasets and applications, including easy-to-use LLM clients such as HuggingChat []. Experimentation with Hugging Face's "sandbox" environment is convenient for users who wish to code, and it's possible to create UI-based clients like ChatBS and share them with the community. Because of the many models available via the platform, at this time (mid 2024) Hugging Face would seem to be an ideal place to host a multi-LLM ChatBS-like question answering client. Further investigation is required to determine if the platform provides access to the resources needed to replicate ChatBS's relationship discovery and entity identification service.

The widespread popularity of ChatBS after its introduction in early 2023 led to a surge in research and publications focusing on various forms of LLM fact checking; of most interest to the ChatBS creators are those that demonstrate the use of ontologies and knowledge graphs as a basis for fact. For example, Gautum [6] presents FactGenius, an approach to fact-checking that combines zero-shot prompting of LLMs with fuzzy text matching on KGs, using DBpedia as an example. Allemang and Sequeda [3] describe an ontology-driven approach to fine-tuning the accuracy SPARQL queries used in a class of question answering systems that combine KGs with LLMs. Liu et.al. [10] describes a novel KG-based approach that does not depend on generated responses, but rather creates a "judge model" to estimate the correctness of the answers given by the LLM. Giarelis, et.al. [7] describe a unified LMM/KG framework intended to provide fact-checking "assistant" capabilities to public deliberation platforms.

6 CONCLUSION/FUTURE WORK

ChatBS's utility as a playground for LLM experimentation could be greatly expanded by adding additional, user-selectable models. The ChatBS team plans to include the recently-released Llama 3 model during mid-2024, with other LLM models to follow.

ChatBS's LLM-based relationship discovery and entity linking service, ChatBS-REST, has proven to be extremely efficient and effective, especially as the core LLM models it is based on have themselves improved. We look forward to expanding the capabilities and applications of ChatBS-REST, in particular by broadening the set of reference knowledge graphs it uses for entity linking beyond Wikidata.

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